

Rapport TP CCNA Security

Infrastructure multi-entreprises avec VLAN, routage OSPFv2 et durcissement sécurité

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Environnement : Cisco Packet Tracer

Équipements : 3 routeurs Cisco 2911, 6 switchs Cisco 2960, 6 PC

1. Objectif

L'objectif du projet est de construire une infrastructure réseau composée de trois entreprises reliées par un backbone WAN en OSPFv2. Chaque entreprise dispose de VLANs dédiés aux étudiants, aux enseignants et à l'administration. Le projet répond au TP demandé et ajoute une couche de durcissement sécurité afin d'obtenir un rendu plus professionnel.

2. Architecture cible

L'infrastructure repose sur un triangle WAN entre R1, R2 et R3. Chaque routeur dessert une entreprise locale via un lien trunk vers un switch S1, lui-même relié à un switch S2. Les VLANs utilisateurs sont routés par sous-interfaces sur les routeurs.

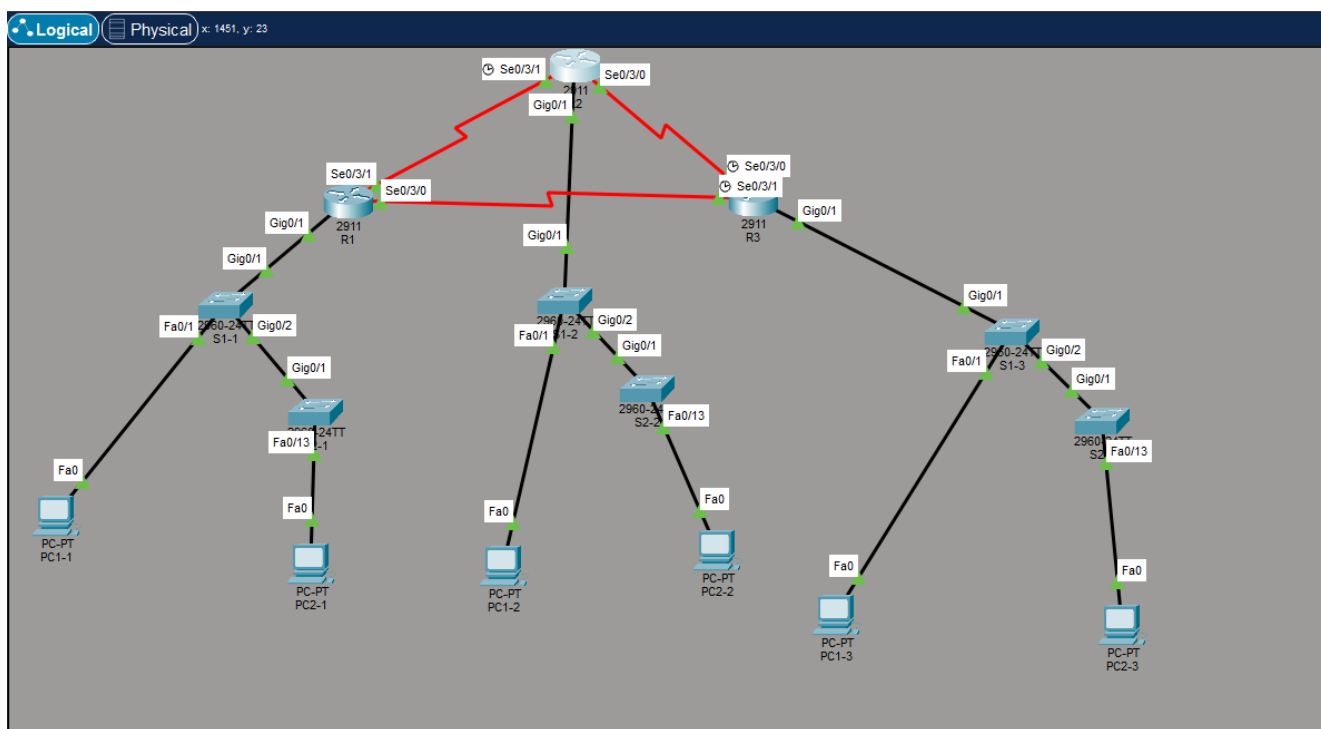


Figure 1 - Topologie finale Packet Tracer.

3. Plan d'adressage synthétique

Zone	VLAN	Réseau	Passerelle	PC
Entreprise 1	10	172.31.192.0/24	172.31.192.254	PC1-1 = 172.31.192.1
Entreprise 1	40	172.31.193.0/24	172.31.193.254	PC2-1 = 172.31.193.1
Entreprise 2	20	192.168.100.0/26	192.168.100.62	PC1-2 = 192.168.100.1
Entreprise 2	50	192.168.100.64/26	192.168.100.126	PC2-2 = 192.168.100.65

Entreprise 3	30	172.18.48.0/24	172.18.48.254	PC1-3 = 172.18.48.1
Entreprise 3	60	172.18.49.0/24	172.18.49.254	PC2-3 = 172.18.49.1

4. Routage OSPFv2

OSPFv2 est configuré en area 0. Les trois routeurs établissent leurs relations de voisinage en état FULL. Les routes distantes apparaissent dans les tables de routage avec le code O, ce qui valide la propagation dynamique.

```
R2#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 0 FULL/- 00:00:39 10.0.0.1 Serial0/3/0
1.1.1.1 0 FULL/- 00:00:39 10.0.0.6 Serial0/3/1
R2#show ip route ospf
10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O 10.0.0.8 [110/128] via 10.0.0.6, 00:03:06, Serial0/3/1
O 10.0.0.8 [110/128] via 10.0.0.1, 00:03:06, Serial0/3/0
O 172.18.0.0/24 is subnetted, 2 subnets
O 172.18.48.0 [110/65] via 10.0.0.1, 00:03:06, Serial0/3/0
O 172.18.49.0 [110/65] via 10.0.0.1, 00:03:06, Serial0/3/0
O 172.31.0.0/24 is subnetted, 2 subnets
O 172.31.192.0 [110/65] via 10.0.0.6, 00:03:06, Serial0/3/1
O 172.31.193.0 [110/65] via 10.0.0.6, 00:03:06, Serial0/3/1
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C 10.0.0.0/30 is directly connected, Serial0/3/0
L 10.0.0.2/32 is directly connected, Serial0/3/0
C 10.0.0.4/30 is directly connected, Serial0/3/1
L 10.0.0.5/32 is directly connected, Serial0/3/1
O 10.0.0.8/30 [110/128] via 10.0.0.6, 00:03:06, Serial0/3/1
O 10.0.0.8/30 [110/128] via 10.0.0.1, 00:03:06, Serial0/3/0
O 172.18.0.0/24 is subnetted, 2 subnets
O 172.18.48.0/24 [110/65] via 10.0.0.1, 00:03:06, Serial0/3/0
O 172.18.49.0/24 [110/65] via 10.0.0.1, 00:03:06, Serial0/3/0
O 172.31.0.0/24 is subnetted, 2 subnets
O 172.31.192.0/24 [110/65] via 10.0.0.6, 00:03:06, Serial0/3/1
R2#show ip protocols
*
* Invalid input detected at '^' marker.
R2#show ip protocols
Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 2.2.2.2
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4
Routing for Networks:
10.0.0.0 0.0.0.3 area 0
10.0.0.4 0.0.0.3 area 0
192.168.100.0 0.0.0.63 area 0
192.168.100.64 0.0.0.63 area 0
Passive Interface(s):
Vlan1
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/2
GigabitEthernet0/1.20
GigabitEthernet0/1.99
GigabitEthernet0/1.20
GigabitEthernet0/1.99
```

Figure 2 - R1 : voisins OSPF, routes OSPF et interfaces passives.

```
172.18.0.0/24 is subnetted, 2 subnets
O 172.18.48.0 [110/65] via 10.0.0.1, 00:03:06, Serial0/3/0
O 172.18.49.0 [110/65] via 10.0.0.1, 00:03:06, Serial0/3/0
O 172.31.0.0/24 is subnetted, 2 subnets
O 172.31.192.0 [110/65] via 10.0.0.6, 00:03:06, Serial0/3/1
O 172.31.193.0 [110/65] via 10.0.0.6, 00:03:06, Serial0/3/1
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C 10.0.0.0/30 is directly connected, Serial0/3/0
L 10.0.0.2/32 is directly connected, Serial0/3/0
C 10.0.0.4/30 is directly connected, Serial0/3/1
L 10.0.0.5/32 is directly connected, Serial0/3/1
O 10.0.0.8/30 [110/128] via 10.0.0.6, 00:03:06, Serial0/3/1
O 10.0.0.8/30 [110/128] via 10.0.0.1, 00:03:06, Serial0/3/0
O 172.18.0.0/24 is subnetted, 2 subnets
O 172.18.48.0/24 [110/65] via 10.0.0.1, 00:03:06, Serial0/3/0
O 172.18.49.0/24 [110/65] via 10.0.0.1, 00:03:06, Serial0/3/0
O 172.31.0.0/24 is subnetted, 2 subnets
O 172.31.192.0/24 [110/65] via 10.0.0.6, 00:03:06, Serial0/3/1
R2#show ip protocols
*
* Invalid input detected at '^' marker.
R2#show ip protocols
Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 2.2.2.2
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4
Routing for Networks:
10.0.0.0 0.0.0.3 area 0
10.0.0.4 0.0.0.3 area 0
192.168.100.0 0.0.0.63 area 0
192.168.100.64 0.0.0.63 area 0
Passive Interface(s):
Vlan1
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/2
GigabitEthernet0/1.20
GigabitEthernet0/1.99
Routing Information Sources:
Gateway Distance Last Update
1.1.1.1 110 00:03:14
2.2.2.2 110 00:03:14
3.3.3.3 110 00:03:14
Distance: (default is 110)
R2#
```

Figure 3 - R2 : validation OSPF.

```

O 192.168.100.64 [110/65] via 10.0.0.2, 00:03:37, Serial0/3/0

R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C 10.0.0.0/30 is directly connected, Serial0/3/0
L 10.0.0.1/32 is directly connected, Serial0/3/0
O 10.0.0.4/30 [110/128] via 10.0.0.9, 00:03:37, Serial0/3/1
O 10.0.0.8/30 [110/128] via 10.0.0.2, 00:03:37, Serial0/3/0
C 10.0.0.8/30 is directly connected, Serial0/3/1
L 10.0.0.10/32 is directly connected, Serial0/3/1
L 172.18.0.0/16 is variably subnetted, 6 subnets, 2 masks
C 172.18.48.0/24 is directly connected, GigabitEthernet0/1.30
L 172.18.48.254/32 is directly connected, GigabitEthernet0/1.30
C 172.18.49.0/24 is directly connected, GigabitEthernet0/1.60
L 172.18.49.254/32 is directly connected, GigabitEthernet0/1.60

R3#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 3.3.3.3
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.0 0.0.0.0 area 0
    10.0.0.8 0.0.0.0 area 0
    172.18.48.0 0.0.0.255 area 0
    172.18.49.0 0.0.0.255 area 0
  Passive Interface(s):
    Vlan1
    GigabitEthernet0/0
    GigabitEthernet0/1
    GigabitEthernet0/2
    GigabitEthernet0/1.30
    GigabitEthernet0/1.60
    GigabitEthernet0/1.99
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110           00:03:48
    2.2.2.2          110           00:03:48
    3.3.3.3          110           00:03:48
  Distance: (default is 110)

R3#
R3#
R3#
R3#
R3#
R3#
R3#
R3#
R3#
R3#

```

Figure 4 - R3 : validation OSPF.

5. Switching, VLANs et trunks

Les liens routeur-S1 et S1-S2 sont configurés en trunk 802.1Q. Le VLAN natif est le VLAN 99, et seuls les VLANs utiles sont autorisés sur chaque trunk. Le VLAN 999 est utilisé pour isoler les ports inutilisés.

```

ACCES STRICTEMENT RESERVE AUX ADMINISTRATEURS AUTORISES

User Access Verification

Password:

S2-1>show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Gig0/1    on        802.1q         trunking      99

Port      Vlans allowed on trunk
Gig0/1    10,40,99

Port      Vlans allowed and active in management domain
Gig0/1    10,40,99

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    10,40,99

S2-1>show vlan brief

VLAN Name                Status    Ports
-----
1    default              active    Fa0/13, Fa0/14, Fa0/15, Fa0/16
10   ETUDIANTS_10          active    Fa0/17, Fa0/18, Fa0/19, Fa0/20
40   ENSEIGNANTS_40         active    Fa0/21, Fa0/22, Fa0/23, Fa0/24
99   ADMIN_NATIVE           active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
999   PARKING_UNUSED         active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                   Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                   Gig0/2
1002 fddi-default          active
1003 token-ring-default   active
1004 fddinet-default       active
1005 trnet-default        active

S2-1>

```

Figure 5 - S1-1 : trunks et VLANs actifs.

```

ACCES STRICTEMENT RESERVE AUX ADMINISTRATEURS AUTORISES

User Access Verification

Password:

S2-3>show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking     99

Port      Vlans allowed on trunk
Gig0/1    30,60,99

Port      Vlans allowed and active in management domain
Gig0/1    30,60,99

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    30,60,99

S2-3>

```

Figure 6 - S2-1 : trunk avec VLAN natif 99.

```

R1#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.4 0.0.0.3 area 0
    10.0.0.8 0.0.0.3 area 0
    172.31.192.0 0.0.0.255 area 0
    172.31.193.0 0.0.0.255 area 0
  Passive Interface(s):
    Vlan1
    GigabitEthernet0/0
    GigabitEthernet0/1
    GigabitEthernet0/2
    GigabitEthernet0/1.10
    GigabitEthernet0/1.40
    GigabitEthernet0/1.99
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110          00:02:48
    2.2.2.2          110          00:02:48
    3.3.3.3          110          00:02:48
  Distance: (default is 110)

R1#

```

Figure 7 - S2-2 : trunk VLAN 20,50,99.

```

ACCES STRICTEMENT RESERVE AUX ADMINISTRATEURS AUTORISES

User Access Verification

Password:

S2-2>show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    99

Port      Vlans allowed on trunk
Gig0/1    20,50,99

Port      Vlans allowed and active in management domain
Gig0/1    20,50,99

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    20,50,99

S2-2>

```

Figure 8 - S2-3 : trunk VLAN 30,60,99.

6. Tests de connectivité

Les tests ICMP confirment la communication de bout en bout entre les entreprises. Les postes des VLANs distants sont joignables grâce au routage OSPF.

```

C:\>ping 172.18.49.1

Pinging 172.18.49.1 with 32 bytes of data:

Reply from 172.18.49.1: bytes=32 time=10ms TTL=126
Reply from 172.18.49.1: bytes=32 time=1ms TTL=126
Reply from 172.18.49.1: bytes=32 time=5ms TTL=126
Reply from 172.18.49.1: bytes=32 time=6ms TTL=126

Ping statistics for 172.18.49.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 5ms

C:\>

```

Figure 9 - PC1-1 vers PC2-3 : connectivité inter-sites validée.

```

PC2-3
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.31.192.1
Pinging 172.31.192.1 with 32 bytes of data:
Reply from 172.31.192.1: bytes=32 time=10ms TTL=126
Reply from 172.31.192.1: bytes=32 time=4ms TTL=126
Reply from 172.31.192.1: bytes=32 time=1ms TTL=126
Reply from 172.31.192.1: bytes=32 time=1ms TTL=126
Ping statistics for 172.31.192.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 4ms
C:\>
C:\>
C:\>
C:\>ping 172.31.192.1
Pinging 172.31.192.1 with 32 bytes of data:
Reply from 172.31.192.1: bytes=32 time=10ms TTL=126
Reply from 172.31.192.1: bytes=32 time=6ms TTL=126
Reply from 172.31.192.1: bytes=32 time=1ms TTL=126
Reply from 172.31.192.1: bytes=32 time=6ms TTL=126
Ping statistics for 172.31.192.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 5ms
C:\>

```

Figure 10 - PC2-3 vers PC1-1 : connectiv   retour valid  e.

7. Durcissement s  curit  

Mesure	Objectif
SSH + login local	��viter l'administration en clair via Telnet.
VLAN natif 99	��viter l'utilisation du VLAN 1 comme VLAN natif.
VLAN 999 parking	Isoler les ports inutilis��s.
Port-security	Limiter les adresses MAC sur les ports d'acc��s.
Passive-interface OSPF	Ne pas envoyer de Hello OSPF vers les LAN utilisateurs.
OSPF MD5	S��curiser les adjacences OSPF sur les liens WAN.
ACL VTY	Restreindre l'administration aux r��seaux autoris��s.

8. Conclusion

Le projet valide les objectifs du TP : VLANs, trunks, router-on-a-stick, OSPFv2, interfaces passives et connectivit   de bout en bout. Il va plus loin en int  grant des mesures de s  curit   r  alistes : VLAN natif d  di  , VLAN parking, port-security, SSH, ACL VTY et authentification OSPF. L'ensemble constitue un rendu propre, d  montrable et d  fendable.